

REAL-TIME OBJECT DETECTION FOR CAMPUS SECURITY USING YOLOV8

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ABSTRACT

This study implements a real-time object detection system for campus perimeter security using YOLOv8, detecting unauthorized persons and abandoned objects with 91.8% mAP on a custom campus dataset.

INTRODUCTION

Campus security monitoring at educational institutions relies heavily on human operators monitoring CCTV feeds, which is prone to fatigue and missed events. Automated object detection using deep learning can augment security staff by flagging suspicious activities. This study develops a YOLOv8-based detection system integrated with the CCTV infrastructure at CvSU Imus Campus.

METHODOLOGY

YOLOv8-medium was trained on a custom dataset of 3,200 annotated CCTV frames covering: Person, Bag (abandoned), Vehicle, and Restricted Area Intrusion classes. Annotation was done using Roboflow. Training: 100 epochs, image size 640×640, batch size 16, AdamW optimizer. The model was deployed on a Jetson Nano (4GB) connected to 4 IP cameras. Alerts are sent via email and SMS when confidence exceeds 0.7.

RESULTS

YOLOv8-medium achieved mAP@0.5 of 91.8% on the validation set. Person class AP was 95.1%; Abandoned Bag AP was 87.3%. Inference speed averaged 28 FPS on Jetson Nano, meeting real-time requirements. False positive rate was 4.2%. The system was stress-tested over 72 hours with zero crashes.

DISCUSSION

YOLOv8 provides robust real-time detection suitable for campus security deployment. Abandoned bag detection remains challenging due to class imbalance (only 12% of dataset). Nighttime performance degraded to mAP 79% due to insufficient low-light training data. Future work should incorporate night-vision training data and integrate with an access control system.